

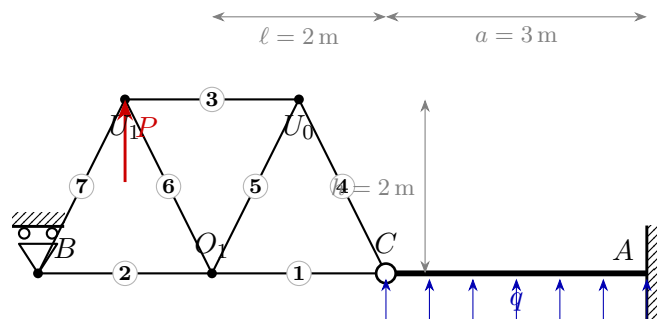
## University of Natural Resources and Life Sciences, Vienna (BOKU)

## VU Mechanik – LAWI 100515 (PI)

Exam **Group A**

Date: 26.06.2026

## Model solution

**1. Static determinacy**Counting criterion  $n = r - (3 + v)$  with  $r$  support reactions and  $v$  hinge conditions:

$$n = r - (3 + v) = 4 - (3 + 1) = 0 \quad \Rightarrow \quad \text{the system is statically determinate.}$$

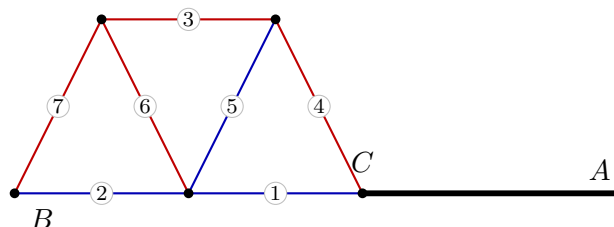
**2. Support reactions and hinge force**

$$A_x = 0 \text{ kN}, \quad A_y = -23 \text{ kN}, \quad M_A = 42 \text{ kNm}, \quad B_x = 0 \text{ kN}, \quad B_y = -15 \text{ kN}$$

$$C_x = 0 \text{ kN}, \quad C_y = 5 \text{ kN}$$

**3. Truss member forces**

| Member    | Member force $S$ [kN] | Type        |
|-----------|-----------------------|-------------|
| 1 (C–O1)  | -2.5                  | compression |
| 2 (O1–O2) | -7.5                  | compression |
| 3 (U0–U1) | 5                     | tension     |
| 4 (C–U0)  | 5.59                  | tension     |
| 5 (U0–O1) | -5.59                 | compression |
| 6 (O1–U1) | 5.59                  | tension     |
| 7 (U1–O2) | 16.77                 | tension     |



red: tension (+)    blue: compression (-)

**4. Section-force diagrams of the bending beam  $A-C$** 